

SOFTWARE REQUIREMENTS SPECIFICATION

Traffic Management System (Simulation)

1. Introduction

1.1 Purpose

The purpose of the Traffic Management System (Simulation) is to simulate basic traffic flow using queues and traffic signals. The system allows vehicles to enter traffic lanes, wait in queues, and move according to traffic signal states and predefined traffic rules.

1.2 Scope

The system will simulate vehicle movement through traffic lanes using queues. It will manage traffic signals, vehicle queues, congestion, and basic traffic statistics such as waiting time, queue length, and vehicles processed. The system will be implemented using C/C++ and will function as a simulation rather than controlling real-world traffic.

1.3 Definitions

Term	Meaning
Vehicle	A simulated vehicle in the system
Queue	A data structure used to maintain waiting vehicles
Traffic Signal	Red, Yellow, or Green signal controlling vehicle movement
Queue Length	Number of vehicles currently waiting
Congestion	A lane having more vehicles than the defined threshold

2. Overall Description

2.1 Product Perspective

The system is a standalone traffic simulation that uses queues to represent vehicles waiting in different lanes. Traffic signals control vehicle movement, while the system updates queues and calculates traffic statistics.

2.2 Main Functions

The system will:

- Add vehicles to the simulation.
- Maintain vehicle queues.
- Manage traffic signal states.
- Process vehicle movement.
- Display traffic information.
- Detect congestion.
- Calculate traffic statistics.

2.3 Users

The system is operated by a user who can interact with the simulation and observe vehicle movement, queues, signals, and statistics.

2.4 Constraints

- Implemented using C/C++.
- Queues are used as the primary data structure.
- The system is a simulation and does not control real-world traffic signals.

2.5 Assumptions

- Vehicles follow the predefined traffic rules.
- Signals operate according to a defined cycle.
- Congestion is determined using a predefined queue-length threshold.

3. Specific Requirements

3.1 Functional Requirements

The functional requirements define the operations and capabilities that the Traffic Management System shall provide.

ID	Functional Requirement
FR1	The system shall allow users to add vehicles to the traffic simulation.
FR2	The system shall maintain vehicles in queues based on traffic lanes and traffic signal states.
FR3	The system shall transition traffic signals between Red, Yellow, and Green states according to a defined signal cycle.
FR4	The system shall implement traffic signal states such as Red, Yellow, and Green.

FR5	The system shall allow vehicles to enter and leave queues according to defined traffic rules.
FR6	The system shall display the current traffic signal state, vehicle queues, and the number of vehicles waiting in each lane.
FR7	The system shall identify lanes experiencing congestion based on the number of vehicles waiting in their queues.
FR8	The system shall calculate basic traffic statistics such as waiting time, queue length, and number of vehicles processed.

3.2 Non-Functional Requirements

The non-functional requirements define the quality characteristics that the system should maintain.

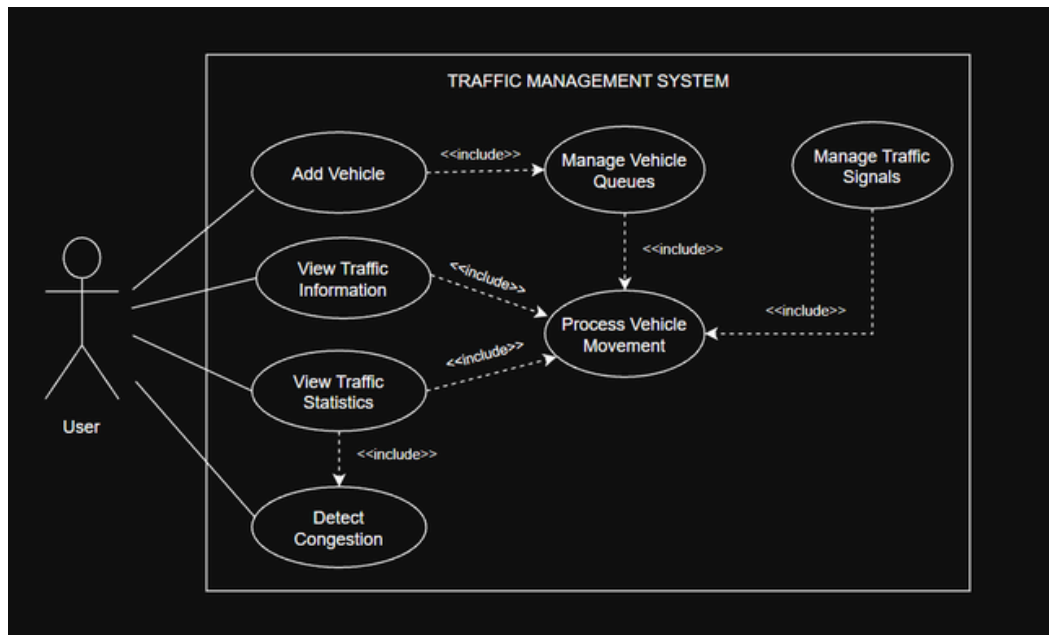
ID	Category	Non-Functional Requirement
NFR1	Performance	The system should process vehicle movements and queue operations efficiently.
NFR2	Usability	The system should provide clear and understandable output showing traffic signals, vehicle queues, and traffic statistics.
NFR3	Reliability	The system should correctly maintain vehicle queues without losing or duplicating vehicles.

3.3 Requirements Summary

The Traffic Management System provides the core functionality required to simulate basic traffic flow using vehicle queues and traffic signals. The functional and non-functional requirements defined above establish the expected system behavior, performance, usability, and reliability. These requirements form the basis for the system's use cases and subsequent design.

4. UML Use Case Diagram

Figure 1: UML Use Case Diagram of the Traffic Management System



5. Security Requirements

5.1 Security Objectives

The security objectives of the Traffic Management System are:

SO1 – Data Integrity

The system shall maintain the integrity of vehicle, queue, signal, and traffic-statistics data throughout the simulation.

SO2 – Input Validation

The system shall ensure that user inputs are validated before being processed to prevent invalid data from affecting the simulation.

5.2 Security Requirements

SR1 – Input Validation

The system shall validate vehicle and simulation inputs before adding or processing them.

SR2 – Queue Data Protection

The system shall prevent invalid operations from causing unintended loss, duplication, or corruption of vehicles in traffic queues.